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Wake-modifying device for a boat

Abstract

A recreational sport boat includes a hull, having starboard and port sides and a transom, and a pair of wake-modifying devices positioned aft of the transom. One of the wake-modifying devices is positioned on a port side of the boat's centerline and another of the wake-modifying devices is positioned on a starboard side of the boat's centerline. Each wake-modifying device includes a plate-like member and at least one downturned surface at a trailing portion of the plate-like member. Each wake-modifying device is pivotable between a non-deployed position and a deployed position about a pivot axis that is horizontal or inclined no more than about 35° from horizontal. When a wake-modifying device is in the deployed position, the downturned surface is lower than it is in the non-deployed position so as to be able to modify the boat's wake.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION (1) This application is a continuation of U.S. patent application Ser. No. 17/152,975, filed Jan. 20, 2021. U.S. patent application Ser. No. 17/152,975 is a continuation of U.S. patent application Ser. No. 17/065,685, filed Oct. 8, 2020, now U.S. Pat. No. 10,899,416. U.S. patent application Ser. No. 17/065,685 is a continuation of Ser. No. 16/360,260, filed Mar. 21, 2019, now U.S. Pat. No. 10,822,055. U.S. patent application Ser. No. 16/360,260 is a continuation of U.S. patent application Ser. No. 15/488,905 filed Apr. 17, 2017,

now U.S. Pat. No. 10,259,534. U.S. patent application Ser. No. 15/488,905 is a continuation of U.S. patent application Ser. No. 14/922,451 filed Oct. 26, 2015, now U.S. Pat. No. 9,643,697. U.S. patent application Ser. No. 14/922,451 is a continuation of U.S. patent application Ser. No. 14/626,249 filed Feb. 19, 2015, now U.S. Pat. No. 9,174,703. U.S. patent application Ser. No. 14/626,249 is a continuation of U.S. patent application Ser. No. 14/458,427 filed Aug. 13, 2014, now U.S. Pat. No. 9,067,644. U.S. patent application Ser. No. 14/458,427 is a continuation of U.S. patent application Ser. No. 14/194,355 filed Feb. 28, 2014, now U.S. Pat. No. 8,833,286. U.S. patent application Ser. No. 14/194,355 claims the benefit under 35 U.S.C. § 119(e) of U.S. Provisional Patent Application No. 61/889,752, filed Oct. 11, 2013, and titled "Wake-Modifying Device for a Boat." The foregoing applications are hereby incorporated by reference in their entireties and are made a part of this specification for all that they disclose.

FIELD OF THE INVENTION

(1) The invention relates to a device for modifying the wake of a boat, as well as a boat equipped with one or more such devices.

BACKGROUND OF THE INVENTION

(2) Recreational sport boats are often used to tow water sports performers such as water skiers, wakeboarders, wake surfers, and the like. The optimal wake depends on which of these water sports a boat is used for, as well as the preferences and skill level of the performer. Water skiers generally prefer a relatively smooth water surface, while wakeboarders and wake surfers desire bigger wakes and wakes with more defined shapes. In recent years, boats have been equipped with various means for modifying the wake of the boat depending on how the boat is being used.

(3) One example of a conventional means used to modify a boat's wake is a trim tab. Trim tabs originally were designed to adjust the trim of a boat. For example, when a boat is overloaded on the port side causing the boat to list to port, a trim tab may be deployed on the port side to cause the boat to return to an even keel. More recently, trim tabs have been used to purposefully modify the wake of a boat. One way to do so is to use one or more trim tabs to lift the stern of the boat. Lifting the stern minimizes the wake of a boat, resulting in a relatively smooth water surface, which is desirable for water skiing. Another way that trim tabs have been used is to increase the displacement of one side of the boat, which increases the size of the wake on the side of the boat with the increased displacement.

(4) FIG. 1 shows the transom **122** of a boat used to tow a water sports performer. Three trim tabs **10**, **20**, **30** known in the art are mounted on the transom **122**: one on the port side, one at the centerline, and one on the starboard side. Each tab is pivotably attached to the transom **122** of the boat by a hinge **11**, **21**, **31**. The port and starboard trim tabs **10**, **30** are flat plates, with inboard and outboard edges bent upwards at a 90° angle. The upwardly-extending portion of each tab extends approximately ¼ inch above the upper surface of the flat portion of the tabs. The center trim tab **20** also is a flat plate, with inboard and outboard edges bent downwards at a 90° angle and a trailing edge bent upwards at a 90° angle. The downwardly-extending portions of the center tab extend approximately ⅛ inch below the lower surface of the flat portion of the tab, and the upwardly-extending portion extends approximately ¼ inch above the upper surface of the flat portion. The edges of the trim tabs **10**, **20**, **30** are bent upwards or downwards to aid in the manufacturing process and the extent to which the edges extend upwards or downwards is preferably minimized.

(5) Each of the aforementioned trim tabs **10**, **20**, **30** is pivotable between a non-deployed position and a deployed position. In FIG. 1, the port trim tab **10** is shown in a deployed position and the starboard trim tab **30** is shown in a non-deployed position. A linear actuator **510** moves each tab between the non-deployed position and the deployed position. As the boat moves through the water, the water flowing under the boat impinges on the deployed trim tab, creating an upward force on the tab. As a result, the portion of the boat where the trim tab is attached is raised. When used for water skiing, the center tab **20** may be deployed to raise the entire stern of the boat and

minimize the wake. To increase the wake of the boat for wake surfing, either one of the port or starboard trim tabs may be deployed. When the port trim tab **10** is deployed, for example, the port side of the boat is raised, causing an increase in displacement on the starboard side of the boat, which increases the size of the starboard wake. The side of the boat with the increased wake is the surf side, which, in this example, is the starboard side. To further increase the size of the wake, ballast may also be added to the surf side of the boat.

(6) In the embodiment shown in FIGS. **1** and **2**, the port and starboard trim tabs **10**, **30** are not perfectly rectangular, but rather have an angled contour along their outboard edges. This is so that the trim tabs do not extend beyond the outer perimeter of a swim platform (not shown) mounted to the transom **122** of the boat, above the trim tabs **10**, **20**, **30**. Alternatively, all of the trim tabs may have a rectangular shape.

(7) Even with the trim tabs described above, wakeboarders and wake surfers desire larger wakes with improved shapes. For surfing in particular, wake surfers desire a wake with a large surfable area. The surfable area is the portion of the wake that pushes the surfer forward. This area generally extends from the curl of the wake to the swim platform.

SUMMARY OF THE INVENTION

(8) In one aspect, the invention relates to a wake-modifying device adapted to be attached to a boat, aft of the boat's transom, on either a port or starboard side of the boat's centerline.

(9) In another aspect, the invention relates to a recreational sport boat including a hull, having starboard and port sides and a transom, and a pair of wake-modifying devices positioned aft of the transom. One of the wake-modifying devices is positioned on a port side of the boat's centerline and another of the wake-modifying devices is positioned on a starboard side of the boat's centerline. Preferably, each wake-modifying device is pivotably attached directly to the transom. Alternatively, one or both of the wake-modifying devices may be attached to other portions of the boat, such as the bottom or sides of the hull or a swim platform.

(10) Each wake-modifying device includes a plate-like member and at least one downturned surface at a trailing portion of the plate-like member. Each wake-modifying device is pivotable between a non-deployed position and a deployed position about a pivot axis that is horizontal or inclined no more than about 35° from horizontal. In the deployed position, the downturned surface is lower than it is in the non-deployed position so as to be able to modify the boat's wake.

(11) The plate-like member and the downturned surface of each wake-modifying device may be an integral piece or separate pieces joined together. The downturned surface may be at a trailing edge of the plate-like member, for example, when they are an integral piece. Or the downturned surface may be inward of the trailing edge, for example, when the downturned surface is a separate piece attached to a lower surface of the plate-like member.

(12) Preferably, an angle between the downturned surface of each wake-modifying device and a lower surface of a central portion of the plate-like member is between about 120° to about 135°, and the downturned surface extends between about 1 inch to about 2¼ inches below the lower surface of the plate-like member. The downturned surface may be oriented such that it intersects the plate-like member along a line that is generally parallel to the pivot axis, or along a line that is at an oblique angle relative to the pivot axis. The downturned surface and the plate-like member need not intersect along a straight line, and may instead intersect along a curved line.

(13) Preferably, each wake-modifying device includes not one but two (or more) downturned surfaces at a trailing portion of the plate-like member. A first one of the downturned surfaces may intersect the plate-like member along a line that is generally parallel to the pivot axis, and a second one of the downturned surfaces may intersect the plate-like member along a line that is at an obtuse angle relative to the line along which the first downturned surface and the plate-like member intersect. Preferably, the obtuse angle is between about 135° to about 150°, and the first downturned surface is outboard of the second downturned surface.

(14) Each wake-modifying device may also include at least one upturned surface at an outboard

portion of the plate-like member, between the downturned surface and the pivot axis. The plate-like member and the upturned surface may be an integral piece or separate pieces joined together, and the upturned surface may be at an outboard edge of the plate-like member or inward of the outboard edge. An angle between the upturned surface and an upper surface of a central portion of the plate-like member preferably is between about 30° to about 150°, and more preferably is about 90°. The upturned surface preferably extends at least about 1 inch above an upper surface of the plate-like member.

(15) At least one of the wake-modifying devices may include another downturned surface at an inboard portion of the plate-like member, between the first downturned surface and the pivot axis. An angle between this downturned surface and the lower surface of a central portion of the plate-like member preferably is between about 30° to about 150°, and more preferably is about 90°. The plate-like member and this downturned surface may be an integral piece or separate pieces joined together, and the downturned surface may be at an inboard edge of the plate-like member or inward of the inboard edge.

(16) At least one of the wake-modifying devices may also include at least one fin attached to the lower surface of the plate-like member. An angle between the fin and the lower surface of the plate-like member preferably is between about 30° to about 150°, and more preferably is about 90°. The fin preferably extends at least about 1 inch below the lower surface of the plate-like member. The fin and the plate-like member preferably intersect along a line that is at an angle between about 15° and about 75° relative to the pivot axis, and more preferably between about 30° and about 60° relative to the pivot axis, and extends in a direction aft and outboard from the pivot axis. In some cases, it may be desirable to have at least two fins, which preferably are parallel to each other.

(17) Each wake-modifying device preferably is capable of assuming multiple deployed positions. In each different deployed position the wake-modifying device is pivoted downwardly at a different angle relative to the non-deployed position. The boat may include a pair of linear actuators, each operable to move a respective one of the wake-modifying devices between its non-deployed position and its deployed position. The boat also preferably includes an operator station with a controller configured to control the operation of each linear actuator.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) FIG. 1 shows a transom of a boat equipped with prior art trim tabs.
- (2) FIG. 2 is a detailed view the starboard trim tab shown in FIG. 1.
- (3) FIG. 3 shows a boat including a pair of wake-modifying devices according to a first preferred embodiment of the invention.
- (4) FIG. 4 is a stern view of the boat shown in FIG. 3.
- (5) FIG. 5 is a detailed view of the port wake-modifying device shown in FIG. 3.
- (6) FIG. 6 is a detailed view of the bottom of the port wake-modifying device shown in FIG. 3.
- (7) FIG. 7 is a perspective view of the port wake-modifying device shown in FIG. 3, detached from the boat and shown here from the inboard side.
- (8) FIG. 8 is a perspective view of the port wake-modifying device shown in FIG. 7, shown here from the outboard side.
- (9) FIG. 9 is a perspective view of the port wake-modifying device shown in FIG. 7, turned upside down and shown here from the outboard side.
- (10) FIG. 10 is an outboard elevation view of the port wake-modifying device shown in FIG. 7.
- (11) FIG. 11 is an inboard elevation view of the port wake-modifying device shown in FIG. 7.
- (12) FIG. 12 is an aft elevation view of the port wake-modifying device shown in FIG. 7.
- (13) FIG. 13 is a top elevation view of the port wake-modifying device shown in FIG. 7.

(14) FIG. 14 is a foreside elevation view of the port wake-modifying device shown in FIG. 7.

(15) FIG. 15 is a bottom elevation view of the port wake-modifying device shown in FIG. 7.

(16) FIG. 16 is a port side view of the boat shown in FIG. 3, with the port wake-modifying device in a non-deployed position.

(17) FIG. 17 is a port side view of the boat shown in FIG. 3, with the port wake-modifying device in a deployed position.

(18) FIG. 18 is a manufacturing view of the port wake-modifying device shown in FIG. 7.

(19) FIG. 19 is a perspective view of the starboard wake-modifying device shown in FIG. 3, detached from the boat and shown here from the outboard side.

(20) FIG. 20 is a perspective view of the starboard wake-modifying device shown in FIG. 19, shown here from the inboard side.

(21) FIG. 21 is a perspective view of the starboard wake-modifying device shown in FIG. 19, turned upside down and shown here from the outboard side.

(22) FIG. 22 is an inboard elevation view of the starboard wake-modifying device shown in FIG. 19.

(23) FIG. 23 is an outboard elevation view of the starboard wake-modifying device shown in FIG. 19.

(24) FIG. 24 is an aft elevation view of the starboard wake-modifying device shown in FIG. 19.

(25) FIG. 25 is a top elevation view of the starboard wake-modifying device shown in FIG. 19.

(26) FIG. 26 is a foreside elevation view of the starboard wake-modifying device shown in FIG. 19.

(27) FIG. 27 is a bottom elevation view of the starboard wake-modifying device shown in FIG. 19.

(28) FIG. 28 is a starboard side view of the boat shown in FIG. 3, with the starboard wake-modifying device in a non-deployed position.

(29) FIG. 29 is a starboard side view of the boat shown in FIG. 3, with the starboard wake-modifying device in a deployed position.

(30) FIG. 30 is a manufacturing view of the starboard wake-modifying device shown in FIG. 19.

(31) FIG. 31 is a stern view of a boat including a pair of wake-modifying devices according to a second preferred embodiment of the invention.

(32) FIG. 32 is a detailed view of the port wake-modifying device shown in FIG. 31.

(33) FIG. 33 is a detailed view of the bottom of the port wake-modifying device shown in FIG. 31.

(34) FIG. 34 is a perspective view of the port wake-modifying device shown in FIG. 31, detached from the boat and shown here from the inboard side.

(35) FIG. 35 is a perspective view of the port wake-modifying device shown in FIG. 34, shown here from the outboard side.

(36) FIG. 36 is a perspective view of the port wake-modifying device shown in FIG. 34, turned upside down and shown here from the outboard side.

(37) FIG. 37 is an outboard elevation view of the port wake-modifying device shown in FIG. 34.

(38) FIG. 38 is an inboard elevation view of the port wake-modifying device shown in FIG. 34.

(39) FIG. 39 is an aft elevation view of the port wake-modifying device shown in FIG. 34.

(40) FIG. 40 is a top elevation view of the port wake-modifying device shown in FIG. 34.

(41) FIG. 41 is a foreside elevation view of the port wake-modifying device shown in FIG. 34.

(42) FIG. 42 is a bottom elevation view of the port wake-modifying device shown in FIG. 34.

(43) FIG. 43 is a port side view of the boat shown in FIG. 31, with the port wake-modifying device in a non-deployed position.

(44) FIG. 44 is a port side view of the boat shown in FIG. 31, with the port wake-modifying device in a deployed position.

(45) FIG. 45 is a manufacturing view of the port wake-modifying device shown in FIG. 34.

(46) FIG. 46 is a perspective view of the starboard wake-modifying device shown in FIG. 31, detached from the boat and shown here from the outboard side.

(47) FIG. 47 is a perspective view of the starboard wake-modifying device shown in FIG. 46, shown here from the inboard side.

(48) FIG. 48 is a perspective view of the starboard wake-modifying device shown in FIG. 46, turned upside down and shown here from the outboard side.

(49) FIG. 49 is an inboard elevation view of the starboard wake-modifying device shown in FIG. 46.

(50) FIG. 50 is an outboard elevation view of the starboard wake-modifying device shown in FIG. 46.

(51) FIG. 51 is an aft elevation view of the starboard wake-modifying device shown in FIG. 46.

(52) FIG. 52 is a top elevation view of the starboard wake-modifying device shown in FIG. 46.

(53) FIG. 53 is a foreside elevation view of the starboard wake-modifying device shown in FIG. 46.

(54) FIG. 54 is a bottom elevation view of the starboard wake-modifying device shown in FIG. 46.

(55) FIG. 55 is a starboard side view of the boat shown in FIG. 31, with the starboard wake-modifying device in a non-deployed position.

(56) FIG. 56 is a starboard side view of the boat shown in FIG. 31, with the starboard wake-modifying device in a deployed position.

(57) FIG. 57 is a manufacturing view of the starboard wake-modifying device shown in FIG. 46.

(58) FIG. 58 is a stern view of a boat including a pair of wake-modifying devices according to a third preferred embodiment of the invention.

(59) FIG. 59 is a detailed view of the port wake-modifying device shown in FIG. 58.

(60) FIG. 60 is a detailed view of the bottom of the port wake-modifying device shown in FIG. 58.

(61) FIG. 61 is a perspective view of the port wake-modifying device shown in FIG. 58, detached from the boat and shown here from the inboard side.

(62) FIG. 62 is a perspective view of the port wake-modifying device shown in FIG. 61, shown here from the outboard side.

(63) FIG. 63 is a perspective view of the port wake-modifying device shown in FIG. 61, turned upside down and shown here from the outboard side.

(64) FIG. 64 is an outboard elevation view of the port wake-modifying device shown in FIG. 61.

(65) FIG. 65 is an inboard elevation view of the port wake-modifying device shown in FIG. 61.

(66) FIG. 66 is an aft elevation view of the port wake-modifying device shown in FIG. 61.

(67) FIG. 67 is a top elevation view of the port wake-modifying device shown in FIG. 61.

(68) FIG. 68 is a foreside elevation view of the port wake-modifying device shown in FIG. 61.

(69) FIG. 69 is a bottom elevation view of the port wake-modifying device shown in FIG. 61.

(70) FIG. 70 is a port side view of the boat shown in FIG. 58, with the port wake-modifying device in a non-deployed position.

(71) FIG. 71 is a port side view of the boat shown in FIG. 58, with the port wake-modifying device in a deployed position.

(72) FIG. 72 is a manufacturing view of the port wake-modifying device shown in FIG. 61.

(73) FIG. 73 is a perspective view of the starboard wake-modifying device shown in FIG. 58, detached from the boat and shown here from the outboard side.

(74) FIG. 74 is a perspective view of the starboard wake-modifying device shown in FIG. 73, shown here from the inboard side.

(75) FIG. 75 is a perspective view of the starboard wake-modifying device shown in FIG. 73, turned upside down and shown here from the outboard side.

(76) FIG. 76 is an inboard elevation view of the starboard wake-modifying device shown in FIG. 73.

(77) FIG. 77 is an outboard elevation view of the starboard wake-modifying device shown in FIG. 73.

(78) FIG. 78 is an aft elevation view of the starboard wake-modifying device shown in FIG. 73.

- (79) FIG. **79** is a top elevation view of the starboard wake-modifying device shown in FIG. **73**.
(80) FIG. **80** is a foreside elevation view of the starboard wake-modifying device shown in FIG. **73**.
(81) FIG. **81** is a bottom elevation view of the starboard wake-modifying device shown in FIG. **73**.
(82) FIG. **82** is a starboard side view of the boat shown in FIG. **58**, with the starboard wake-modifying device in a non-deployed position.
(83) FIG. **83** is a starboard side view of the boat shown in FIG. **58**, with the starboard wake-modifying device in a deployed position.
(84) FIG. **84** is a manufacturing view of the starboard wake-modifying device shown in FIG. **73**.
(85) FIG. **85** shows an exemplary control screen for operating the wake-modifying devices shown in the previous figures.
(86) FIG. **86** shows an edit screen accessed from the control screen shown in FIG. **85**.
(87) FIG. **87** shows another exemplary control screen for operating the wake-modifying devices shown in the previous figures.

(88) FIG. **88** shows a selection screen accessed from the control screen shown in FIG. **87**.

DETAILED DESCRIPTION OF THE PREFERRED EMBODIMENTS

(89) Exemplary preferred embodiments of the invention will now be described with reference to the accompanying figures. Like reference numerals refer to the same or similar elements throughout the figures and description.

First Embodiment

- (90) FIGS. **3** and **4** show a boat **100** equipped with a pair of wake-modifying devices **111**, **112** in accordance with a first preferred embodiment of the invention. The boat hull **120** includes a bow **121**, a transom **122**, and port and starboard sides **123**, **124**. Within the boat's interior is a control console **140** for operating the boat **100**. The boat **100** is driven by a single inboard motor (not shown) connected to a left-handed propeller **170**. However, the wake-modifying devices **111**, **112** can be utilized with other types of boats and propulsion systems, including but not limited to right-handed propellers, outboard motors, sterndrives, and the like.
- (91) The boat **100** has a centerline **150** running down the center of the boat, halfway between the port and starboard sides **123**, **124**. A conventional trim tab **20** is pivotably attached to the transom **122** along the centerline **150**. The wake-modifying devices **111**, **112** are pivotably attached to the transom **122** on port and starboard sides of the centerline **150**, respectively.
- (92) FIGS. **5-18** are detailed views of the wake-modifying device **111** on the port side of the boat **100**. Similarly, FIGS. **19-30** are detailed views of the wake-modifying device **112** on the starboard side of the boat **100**. While the port and starboard wake modifying-devices **111**, **112** have many similar features in this embodiment, the port and starboard wake-modifying devices **111**, **112** differ with respect to one downturned surface **330**, which the port wake-modifying device **111** includes and the starboard wake-modifying device **112** lacks, as will be discussed further below. Except where noted otherwise, the discussion of the wake-modifying devices applies equally to both the port and starboard wake-modifying devices **111**, **112**.
- (93) Each wake-modifying device **111**, **112** includes a plate-like member **200** that is pivotably attached to the transom **122** of the boat **100**. The plate-like member **200** pivots about a pivot axis **210** to move between a non-deployed position and a deployed position. In this embodiment, the pivot axis **210** is a hinge and is flush with the transom **122** of the boat **100**. Here, the hinge is a piano hinge that is welded to a leading portion **L** of the plate-like member **200** and attached to the transom of the boat **100** using screws. However, any suitable pivotable connection may be used and it may be affixed to the wake-modifying device **111**, **112** and transom **122** of the boat **100** using any suitable means, including but not limited to bolts, screws, rivets, welding, and epoxy. In addition, the wake modifying device **111**, **112** may be attached to the transom **122** such that the pivot axis **210** is not flush with the transom **122**, for example, the pivot axis may be spaced further aft of the transom **122**. The wake-modifying devices **111**, **112** also may be attached to portions of the boat

other than the transom **122**. For example, the wake-modifying devices **111**, **112** could be attached to the bottom of the hull **120**, to the port and starboard sides **123**, **124** of the hull **120**, or to a swim platform (not shown). The pivot axis **210** preferably is parallel to the transom **122**, but it may be oriented at an oblique angle relative to the transom **122** so long as the wake-modifying device **111**, **112** provides an upward force on the boat **100** as the boat **100** travels forward through the water. (94) In the embodiment shown, the pivot axis **210** is parallel to the deadrise (the angle of the hull from the keel to the chine at the transom **122**) of the boat **100**. But the pivot axis **210** may instead be at an angle relative to the deadrise. Some boats, for example, have little or no deadrise. In such cases, it may be advantageous to orient the pivot axis **210** at an angle relative to the deadrise. Preferably, the pivot axis is inclined no more than about 35° from horizontal, more preferably no more than about 20° from horizontal, and most preferably no more than about 15° from horizontal. This inclination is preferably in the direction from the chine to the keel. Preferably, the pivot axis is inclined no more than about 15° more than the deadrise.

(95) The plate-like member **200** has a trailing portion T that is aft of the leading portion L. The trailing portion T is the aft half of the plate-like member **200**, and the leading portion L is the forward half of the plate-like member **200**. The plate-like member **200** also has an inboard portion I and an outboard portion O. The inboard portion I is the inboard half of the plate-like member, and the outboard portion O is the outboard half of the plate-like member. Thus, the plate-like member may be divided into quadrants as shown in FIGS. **13** and **25**: a leading, inboard portion L, I; a leading, outboard portion L, O; a trailing, inboard portion T, I; and a trailing, outboard portion T, O. (96) In the embodiment shown, there are two downturned surfaces **310**, **320** at the trailing portion T of the plate-like member **200**. The first downturned surface **310** intersects the plate-like member **200** along a line that is generally parallel to the pivot axis **210**. The second downturned surface **320** intersects the plate-like member **200** along a line that is oriented at an angle α relative to the pivot axis **210**. In this embodiment, the first downturned surface **310** is outboard of the second downturned surface **320**, which is at the trailing, inboard portion T, I of the plate-like member **200**. The second downturned surface **320** may extend into adjacent quadrants without deviating from the scope of the invention. The first and second downturned surfaces **310**, **320** preferably are at the edge of the plate-like member **200**, but they may be inward of the edge.

(97) The inventors believe that the combination of the plate-like member **200** and one or both of the downturned surfaces **310**, **320** improves the size and shape of the wake. The side of the boat **100** with the desirable wake is referred to as the surf side. The surf side is the side of the boat **100** opposite a deployed wake-modifying device. The side with the deployed wake-modifying device is referred to as the non-surf side.

(98) As the boat **100** moves through the water, the hull displaces water both downward under the hull **120** and outward of the sides **123**, **124** of the hull **120**. This creates a cavity immediately behind the boat **100**. The displaced water recovers behind the boat **100** to fill the cavity. As the displaced water recovers, the water converges from under the boat **100** and from the sides **123**, **124** of the hull **120**. When the convergence occurs with sufficient force, it creates a v-shaped wave crest or “rooster tail” at the point of convergence. This v-shaped crest then propagates outward behind the boat **100** creating a wake that is suitable for wakeboarding, wake surfing, and the like. When one of the wake-modifying devices **111**, **112** is in the deployed position, the downturned surfaces **310**, **320** direct the water sharply downward. This sharp redirection of water results in an additional upward force to roll the boat **100** toward the surf side to a greater degree than the prior art trim tabs discussed above and shown in FIGS. **1** and **2**. As a result, the boat **100** displaces more water on the surf side creating a more desirable wake for surfing. The inventors have found that shifting the point of convergence aft of the transom **122** and toward the non-surf side will also improve the wake on the surf side. The inventors believe that creating a trough in the water behind the deployed wake-modifying device and delaying the water from recovering from the non-surf side shifts the convergence point aft of the transom **122** and toward the non-surf side. The downturned surfaces

310, 320 are believed to contribute to both of these actions. The angles of the downturned surfaces and the size of the plate contribute to the timing of water recovery to increase the size of the wake on the surf side as a result of wave superposition.

(99) The downturned surface **310, 320** should extend far enough in a downward direction to cause redirection of the water. The downturned surface **310, 320** should also be short enough that the downturned surface does not interact with the water when in the non-deployed position. Preferably, the downturned surface **310, 320** extends from about 1 inch to about $2\frac{1}{4}$ inches below a lower surface **220** of the plate-like member **200**, and more preferably about $1\frac{1}{2}$ inches below the lower surface **220** of the plate-like member **200**. The downturned surface **310, 320** forms an angle β with a lower surface **220** of a central portion of the plate-like member **200**. The inventors have found that this angle β should be sufficient to redirect the water, but not so sharp as to result in excessive force on the wake-modifying device **111, 112**. Preferably, the angle β between the downturned surface **310, 320** the lower surface **220** of the central portion of the plate-like member **200** is between about 120° and about 135° .

(100) The water converging behind the boat **100** from the sides **123, 124** of the hull **120** forms an angle with the sides of the hull **123, 124**. The inventors have found that orienting a downturned surface **320** to intersect this angle improves the wake on the surf side. Accordingly, the line where the second downturned surface **320** intersects the plate-like member **200** is oriented at an angle α relative to the pivot axis **210**. This angle α preferably is perpendicular to the angle formed between the recovering water and the side **123, 124** of the hull **120**. The angle α preferably is between about 30° to about 45° .

(101) In the embodiment shown, the wake-modifying device **111, 112** has two upturned surfaces **410, 420** between the first downturned surface **310** and the pivot axis **210**. These upturned surfaces **410, 420** are at the outboard portion O of the plate-like member **200**, preferably at the edge of the plate-like member, but they may be inward of the edge. An angle γ between the upturned surfaces **410, 420** and an upper surface **230** of the central portion of the plate-like member **200** preferably is between about 30° to about 150° , and more preferably is about 90° . The inventors believe that these upturned surfaces **410, 420** delay the water on the non-surf side from converging behind the boat **100** and further shift the point of convergence aft of the transom **122** and toward the non-surf side. The upturned surfaces **410, 420** should extend far enough in an upward direction to delay the water. The upturned surfaces **410, 420** preferably extend at least about 1 inch above the upper surface **230** of the plate-like member **200**, more preferably at least about 2 inches above the upper surface **230** of the plate-like member **200**, and even more preferably at least about 2.5 inches above the upper surface **230** of the plate-like member **200**. In this embodiment, the first upturned surface **410** intersects the plate-like member along a line that is generally parallel to either the port side **123** of the hull **120** or the starboard side **124** of the hull **120**. The second upturned surface **420** is positioned between the first upturned surface **410** and the first downturned surface **310** and intersects the plate-like member along a line that is oriented at an oblique angle δ with respect to the pivot axis **210**. The angle δ preferably is between about 60° to about 90° , and more preferably is about 75° .

(102) A third downturned surface **330** between the second downturned surface **320** and the pivot axis **210** can further improve the wake on the surf-side. An angle ϵ between the third downturned surface **330** and the lower surface **220** of the central portion of the plate-like member **200** preferably is between about 30° to about 150° , and more preferably is about 90° . The inventors believe this third downturned surface **330** further delays the water on the non-surf side from converging with the water on the surf side. In the embodiment shown, the third downturned surface **330** is positioned along the inboard portion I of the plate-like member **200** and intersects the plate-like member **200** along a line that is generally perpendicular to the pivot axis **210**. The third downturned surface **330** preferably is at an inboard edge of the plate-like member, but it may be inward of the edge. Similar to the other downturned surfaces **310, 320**, the third downturned

surface **330** should extend far enough in a downward direction to delay or redirect the water. As with the other downturned surfaces **310**, **320**, the third downturned surface **330** is preferably short enough that the third downturned surface **330** does not interact (or at least minimizes interaction) with the water when in the non-deployed position. The third downturned surface **330** preferably extends between about ½ inch to about 3 inches below the lower surface **220** of the plate-like member **200**, and more preferably extends about 1 inch below the lower surface **220** of the plate-like member **200**.

(103) The boat **100** of this embodiment uses a left-handed propeller **170**, which causes the prop wash to be offset towards the starboard side. To balance the desirability of the surf wakes on both sides of the boat **100**, the port wake-modifying device **111** of this embodiment provides more delay of the water than does the starboard wake-modifying device **112**. In this embodiment, the port wake-modifying device **111** has the third downturned surface **330** while the starboard wake-modifying device **112** does not. However, the third downturned surface **330** may be provided on either the port or starboard wake-modifying device **111**, **112**, both, or neither.

(104) FIG. **16** shows the port wake-modifying device **111** in a non-deployed position, and FIG. **17** shows the port wake-modifying device **111** in a deployed position. Similarly, FIG. **28** shows the starboard wake-modifying device **112** in a non-deployed position, and FIG. **29** shows the starboard wake-modifying device **112** in a deployed position. As water recovers from under the boat, it travels at an upward angle. The angle of water recovery will depend on a number of factors including hull design and operational parameters, for example, speed. In the non-deployed position, the wake-modifying device **111**, **112** preferably is at an upward angle so that it does not interact with the recovering water. The wake-modifying device **111**, **112** is moved from the non-deployed position to a deployed position by pivoting about the pivot axis **210**. In the deployed position, the trailing portion T of the wake-modifying device **111**, **112** is lower than it is in the non-deployed position. The deployed position preferably is even with the bottom of the hull and more preferably extends at a downward angle ζ , as shown in FIGS. **17** and **29**. As discussed above, the wake-modifying device **111**, **112** may be pivotable about an axis that is not horizontal, for example, it may be at an angle parallel to the deadrise. The deployed position is not limited to a single angle ζ , but rather may vary depending upon the preferences of the water sports performer. In this embodiment, the angle ζ is directly proportional to the size of the wake, but as the angle ζ gets larger, the wake begins to break (curl over) closer to the boat **100**, reducing the area on the wake that is desirable for surfing. In this embodiment, the angle ζ preferably is between about 0° and about 12°.

(105) In the embodiment shown, a linear actuator **510** is used to move the wake-modifying device **111**, **112** between the deployed and non-deployed positions. The linear actuator **510** preferably is an electric linear actuator, such as one available from Lenco Marine. One end of the linear actuator **510** is screwed to the transom **122** of the boat **100**. The other end of the linear actuator is connected to a u-shaped bracket **202** by a pin **204**. The u-shaped bracket **202** is then bolted to the plate-like member **200**. Any suitable means may be used to move the wake-modifying device **111**, **112** between the deployed and non-deployed positions, including but not limited to hydraulic linear actuators and mechanical levers.

(106) The size of the wake-modifying device **111**, **112** may be varied depending upon the characteristics of the boat **100** and the desired wake. The lift provided by the wake-modifying device **111**, **112** is generally proportional to the angles α , β , the surface area of the first and second downturned surfaces **310**, **320**, and the surface area of the plate-like member **200**. In this embodiment, the wake-modifying device **111**, **112** is about 17 inches long and about 14.5 inches wide. Preferably, the wake-modifying device **111**, **112** is at least about 10 inches long and at least about 9 inches wide.

(107) Because the wake-modifying device **111**, **112** is used in a marine environment, it preferably is made of materials suitable for that environment. In this embodiment, these materials are primarily

corrosion-resistant metal alloys such as stainless steel. The wake-modifying device **111, 112**, including the plate-like member **200**, preferably should not deform during operation. Preferably, the wake-modifying device **111, 112** will have sufficient rigidity to maintain its shape at all speeds and especially at speeds suitable for surfing (approximately 9 mph to 12 mph). In the preferred embodiment, the wake-modifying device **111, 112** is made from 12 gauge stainless steel plate. Other suitable materials may be used instead, such as wood, plastic, fiber reinforced composites, or other metals including aluminum.

(108) FIGS. **18** and **30** show manufacturing views of the port and starboard wake-modifying devices **111, 112**, respectively. The wake-modifying device **111, 112** may be manufactured by cutting the device from a single plate stock to the desired shape and then bending the downturned and upturned surfaces to the desired angles. In this embodiment, the plate-like member **200**, the downturned surfaces **310, 320, 330**, and the upturned surfaces **410, 420** are an integral piece. Alternatively, some or all of the downturned and upturned surfaces may be formed as separate pieces and then attached to the plate-like member using any suitable joining method such as welding, bolting, riveting, or the like.

Second Embodiment

(109) Boat design plays an important role in establishing the wake shape. Design factors include, for example, the hull design and the weight of the boat. The wake-modifying device **111, 112** preferably is customized based on the boat design in order to produce the desired wake.

(110) As an example of how hull design affects the boat's wake, a first boat having a steeper deadrise than a second boat will typically allow the water to recover closer to the transom of the boat. The shape of the corners between the sides of the hull and the transom also impact the recovery of the water. A boat with smooth corners (e.g., having a radius) will allow the water to recover faster than will a boat with square corners. The wake-modifying devices **111, 112** of the second embodiment are designed to provide greater lift to the boat on the non-surf side and further delay and direct the water on the non-surf side.

(111) FIG. **31** shows a stern view of a boat **100** equipped with wake-modifying devices **111, 112** according to a second preferred embodiment of the invention. The port wake-modifying device **111** of the second embodiment is shown in FIGS. **32-45**. The starboard wake-modifying device **112** of the second embodiment is shown in FIGS. **46-57**.

(112) As with the first embodiment, the port and starboard wake-modifying devices **111, 112** of the second embodiment are not symmetrical with one another. Here, the port wake-modifying device **111** includes two fins **340, 350** attached to the lower surface **220** of the plate-like member **200**. These fins **340, 350** extend at a downward angle η relative to the lower surface **220** of the plate-like member **200**. The downward angle η preferably is between about 30° and about 150° , and more preferably is about 90° . The fins **340, 350** intersect the plate-like member **200** along lines that are oriented at an angle θ relative to the pivot axis **210**. This angle θ preferably is between about 15° and about 75° , and more preferably is between about 30° and about 60° . In the embodiment shown, the fins **340, 350** are parallel to each other and at an angle θ of 60° relative to the pivot axis **210**. The inventors believe these fins **340, 350** redirect the water outboard, further shifting the point of convergence aft from the transom **122** and toward the non-surf side. As with the downturned surfaces **310, 320, 330** and the upturned surfaces **410, 420**, the fins **340, 350** should extend far enough away from the plate-like member **200** to redirect the water flow. The fins **340, 350** preferably should extend at least about 1 inch below the lower surface **220** of the plate-like member **200**, and more preferably should extend at least about 2 inches below the lower surface **220** of the plate-like member **200**. In the embodiment shown, the fins **340, 350** have different lengths. The fins preferably are about 4 inches to about 16 inches long, and more preferably are about 8 inches to about 12 inches long. The fins preferably do not extend beyond the edges of the plate-like member.

(113) In the second embodiment, the third downturned surface **330** intersects the plate-like member along a line that is oriented at an oblique angle ϕ relative to the pivot axis **210**. The angle ϕ

preferably is between about 45° to about 90°, and more preferably is about 70°. All three of the downturned surfaces **310, 320, 330** form an angle α , ϵ with respect to the lower surface **220** of the central portion of the plate-like member **200** of about 135°. The wake-modifying devices **111, 112** of this embodiment preferably are about 17 inches wide and about 16 and ½ inches long.

Third Embodiment

(114) The port and starboard wake-modifying devices **111, 112** in the previous two embodiments are asymmetrical with each other, but the wake-modifying devices **111, 112** can be symmetrical. Symmetrical wake-modifying devices **111, 112** may be used, for example, with heavy boats where the propeller **170** has less of an impact on the boat's wake. The third preferred embodiment shown in FIGS. **58-84** is an example of a symmetrical pair of wake-modifying devices **111, 112**. FIG. **58** shows a stern view of a boat **100** equipped with wake-modifying devices **111, 112** according to a third embodiment of the invention. The port wake-modifying device **111** of the third embodiment is shown in FIGS. **59-72**. The starboard wake-modifying device **112** of the third embodiment is shown in FIGS. **73-84**. In this embodiment, both the port and starboard wake-modifying devices **111, 112** have two fins **340, 350**. In the embodiment shown, the fins **340, 350** are parallel to each other and oriented at an angle θ of 45° relative to the pivot axis **210**.

(115) Control System

(116) Those skilled in the art understand that the weight and displacement of the boat has a significant impact on the size and shape of the wake. As a result, many recreational sport boats that are used for wakeboarding and wake surfing accommodate additional weight or ballast. In particular, many boats are designed to have ballast added to the surf side of the boat to increase the displacement of that side of the boat. This weight may be added by any number of ways known to those skilled in the art. One way is to position more people on the surf side of the boat than on the non-surf side. Another way is to add ballast through the use of ballast bags or ballast sacks. Yet another way to add weight is through ballast tanks installed in the boat. Preferably, two ballast tanks are positioned in the stern of the boat near the bottom of the hull, one on each side of the boat, and a third ballast tank is positioned along the boat's centerline near the bottom of the hull, forward of the two rear ballast tanks. If ballast bags are used in addition to ballast tanks, the ballast bags may be plumbed into the ballast system of the boat. Both the ballast tanks and the ballast bags operate similarly in that water may be pumped into the tank or bag by ballast pumps to add weight. In some boats, both ballast tanks and ballast bags may be used simultaneously. For example, all three ballast tanks may be filled to increase the displacement of the stern of the boat, and a ballast bag on the surf side of the boat may be filled to further increase the displacement on the surf side.

(117) A control system is used to operate the wake-modifying devices **111, 112**. When the wake-modifying devices **111, 112** are used with plumbed-in ballast, the control system preferably controls both the ballast and the wake-modifying devices **111, 112**. This control system preferably includes a controller that controls the linear actuators **510** and the ballast pumps. The controller may be any suitable controller known in the art including a controller comprising a CPU, ROM, and RAM. The control system also includes an input device. In the preferred embodiment, the input device is a touch screen located at the control console **140** of the boat **100**. Also in this embodiment, the controller is co-located with the touch screen. Those skilled in the art will recognize that any suitable input device including buttons, switches, dials, or the like may be used.

(118) An exemplary touch screen **610** is shown in FIG. **85**. This touch screen **610** is shown in a manual mode. In this mode, a user can manually adjust the amount of water in the ballast tanks and manually adjust the percentage of deployment of the wake-modifying devices **111, 112**. To adjust ballast, the user selects the "BALLAST" button **611**. To adjust the deployment of the wake-modifying devices or center trim tab, the user selects the "TABS" button **612**. To adjust the speed of the boat, the user selects the "SET SPEED" button **613**. When one of these buttons is selected, a new screen is displayed that allows the user to adjust the selected parameter. When the user selects the "TABS" button **612**, for example, an "EDIT TABS" screen **620** is displayed as shown in FIG.

86. On this screen **620**, the user may adjust the percentage deployment of the wake-modifying devices **111**, **112** and/or the center trim tab **20**. To deploy the port wake-modifying device **111**, for example, a user swipes his or her finger in direction A in the area **621** until the desired percentage deployment (50 percent in the example shown) is reached. Alternatively, the user may use adjustment arrows **622** to incrementally change the percentage deployment. When the percentage deployment of the port wake-modifying device **111** is changed, the controller drives the linear actuator **510** of the port wake-modifying device **111** to move the port wake-modifying device **111** to the set position. The starboard wake-modifying device **112** and the center trim tab **20** are adjusted in the same manner.

(119) As another option, the wake-modifying devices, ballast, and boat speed may be controlled using user-defined programmed settings. A user can manually set each of the parameters in the manual mode as described above and then save these settings as a user-defined profile. In operation, a touch location **610**, shown in FIG. **85**, may be used to toggle between these user-defined programmed settings. In this embodiment, a user toggles between settings by swiping touch position **614** to the left or right.

(120) In addition to or instead of the foregoing, the control system can include preprogrammed settings established by the boat manufacturer. After one of the preprogrammed settings has been selected by a user, a cruise control screen **630** may be displayed, such as shown in FIG. **87**. The boat may be set to cruise at a specific speed by selecting the "ON" button **632**. If the user desires to switch between preprogrammed settings, the user may select the "SWITCH" button **631**. Selecting the "SWITCH" button **631** causes a pop-up window to be displayed within the control screen **640**, such as shown in FIG. **88**. Using the pop-up window the user may select a different preprogrammed setting. In this embodiment, four preprogrammed settings are shown. The two "mellow" settings ("SURF LEFT MELLOW" **641** and "SURF RIGHT MELLOW" **643**) may be programmed such that the controller drives the linear actuator **510** to deploy the wake-modifying device **111**, **112** on the non-surf side to a relatively shallow deployed position. The two "steep" settings ("SURF LEFT STEEP" **642** and "SURF RIGHT STEEP" **644**) may be programmed to maximize the size of the wake. In these "steep" settings, the controller drives the linear actuator **510** to deploy the wake-modifying device **111**, **112** on the non-surf side to the maximum deployed position. In addition to or instead of a touch screen, other known input devices, such as static buttons **650**, can be used.

(121) The embodiments described and shown herein are examples of preferred embodiments of the present invention and are provided for illustrative purposes only. They are not intended to limit the scope of the invention. Although specific configurations, structures, materials, etc. have been shown and described, such are not limiting. Modifications and variations are contemplated within the scope of the invention, which is to be limited only by the scope of the claims.

Claims

1. A recreational sport boat comprising: a hull including starboard and port sides and a transom; and a pair of wake-modifying devices positioned aft of the transom, one of the wake-modifying devices being positioned on a port side of the boat's centerline and another of the wake-modifying devices being positioned on a starboard side of the boat's centerline, each wake-modifying device being pivotable between a non-deployed position and a deployed position about a pivot axis that is horizontal or inclined no more than about 35° from horizontal, each wake-modifying device including a plate-like member and at least one fin attached to a lower surface of the plate-like member and extending from a central portion of a lower surface of the plate-like member, the fin being angled obliquely relative to the pivot axis, wherein in the deployed position, a trailing portion of the plate-like member is lower than it is in the non-deployed position so as to be able to modify the boat's wake.

2. The recreational sport boat of claim 1, wherein the fin of each wake-modifying device extends in an aft and outboard direction.
 3. The recreational sport boat of claim 1, wherein the fin of each wake-modifying device is angled between about 15° and about 75° relative to the pivot axis.
 4. The recreational sport boat of claim 1, wherein the fin of each wake-modifying device is one of a plurality of fins attached to the lower surface of the plate-like member.
 5. The recreational sport boat of claim 4, wherein at least two fins of the plurality of fins are parallel to each other.
 6. The recreational sport boat of claim 1, wherein an angle between the fin and the lower surface of a central portion of the plate-like member is about 90°.
 7. The recreational sport boat of claim 1, wherein an edge of the trailing portion of the plate-like member of each wake-modifying device is downturned.
 8. A wake-modifying device configured to generate a surf wake behind a boat, the wake-modifying device comprising: a plate-like member adapted to be pivotally mounted aft of a transom of the boat on either a port side of the boat's centerline or a starboard side of the boat's centerline; and a fin attached to a lower surface of a central portion of the plate-like member, the fin being angled obliquely relative to a leading edge of the plate-like member.
 9. The wake-modifying device of claim 8, wherein the fin is angled between about 15° and about 75° relative to the leading edge of the plate-like member.
 10. The wake-modifying device of claim 8, wherein an angle between the fin and the lower surface of a central portion of the plate-like member is about 90°.
 11. The wake-modifying device of claim 8, wherein the fin is one of a plurality of fins attached to the lower surface of the plate-like member.
 12. The wake-modifying device of claim 11, wherein the fins are parallel to each other.
 13. The wake-modifying device of claim 8, wherein an edge of the trailing portion of the plate-like member is downturned.
 14. A recreational sport boat comprising: a hull including starboard and port sides and a transom; and a pair of wake-modifying devices positioned aft of the transom, one of the wake-modifying devices being positioned on a port side of the boat's centerline and another of the wake-modifying devices being positioned on a starboard side of the boat's centerline, each wake-modifying device being pivotable between a non-deployed position and a deployed position about a pivot axis that is horizontal or inclined no more than about 35° from horizontal, each wake-modifying device including: a plate-like member having a forward half and an aft half; and a fin attached to a lower surface of a central portion of the plate-like member, the fin extending from the forward half of the plate-like member to the aft half of the plate-like member, wherein in the deployed position, a trailing portion of the plate-like member is lower than it is in the non-deployed position so as to be able to modify the boat's wake.
 15. The recreational sport boat of claim 14, wherein the fin of each wake-modifying device is angled obliquely relative to the pivot axis and extends in an aft and outboard direction.
 16. The recreational sport boat of claim 15, wherein the fin of each wake-modifying device is angled between about 15° and about 75° relative to the pivot axis.
 17. The recreational sport boat of claim 14, wherein the fin of each wake-modifying device is one of a plurality of fins attached to the lower surface of the plate-like member.
 18. The recreational sport boat of claim 17, wherein at least two fins of the plurality of fins are parallel to each other.
 19. The recreational sport boat of claim 14, wherein an angle between the fin and the lower surface of a central portion of the plate-like member is about 90°.
 20. The recreational sport boat of claim 14, wherein an edge of the trailing portion of the plate-like member of each wake-modifying device is downturned.
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